MAAT v0.10: Observable Signatures of Structural Selection in Cosmology

Expansion, Equation of State, Density Fraction, and Growth Proxies

Christof Krieg
MAAT Project – Independent Research
https://maat-research.com

2026

Abstract

MAAT v0.9 established a stable toy FLRW branch for a kinetic structural-selection field. This paper introduces MAAT v0.10, an observable-prediction layer that converts the representative v0.9 trajectory into directly interpretable cosmological proxies.

The construction computes the expansion history $E(z) = H(z)/H_0$, the relative Hubble deviation from an internally normalised ΛCDM reference, the MAAT equation of state $w_{\mathrm{MAAT}}(z)$, the MAAT density fraction $\Omega_{\mathrm{MAAT}}(z)$, and a first growth proxy $f\sigma_8(z)$ based on the growth-index approximation.

For the representative stable v0.9 trajectory, the projection covers $0 \leq z \leq 2.329$ and yields a present MAAT fraction $\Omega_{\mathrm{MAAT},0} = 0.00331$, a present equation of state $w_{\mathrm{MAAT},0} = -0.801$, a maximum relative Hubble deviation $|\Delta H/H|_{\max} = 0.0399$, and a maximum growth-proxy deviation $|\Delta f\sigma_8/f\sigma_8|_{\max} = 0.0336$.

The result is not a fit to cosmological data and not a full perturbation calculation. Its contribution is to make the v0.9 stability branch observationally falsifiable in principle by defining measurable expansion and growth signatures.

1 Purpose and Status

The previous MAAT development established a chain from covariance-driven selection pressures to local fields, gravitational coupling, and a stable toy FLRW branch. Stability alone, however, is not enough. A cosmological model must eventually make observable predictions.

The purpose of MAAT v0.10 is therefore:

> *to convert a stable structural-selection trajectory into observable expansion and growth signatures.*

This paper is deliberately limited in scope. It does not fit supernovae, BAO, CMB, weak-lensing, or redshift-space distortion data. It does not solve the Hubble tension, the S_8 tension, or the dark-energy problem. It defines a reproducible observable layer that future versions can compare against data.

2 Input from v0.9

The input is the representative stable trajectory from the v0.9 FLRW scan. The effective background equation is

$$3H^2 = \rho_m + \rho_r + \rho_\Lambda + \rho_\phi + \rho_{\mathrm{MAAT}}. \tag{1}$$

The MAAT component is described by an effective density and pressure,

$$w_{\mathrm{MAAT}}(z) = \frac{p_{\mathrm{MAAT}}(z)}{\rho_{\mathrm{MAAT}}(z)}. \tag{2}$$

The trajectory is not interpreted as a calibrated physical cosmology. Instead, the slice closest to $a = 1$ is used as the effective present epoch:

$$a_0 = 0.998777, \qquad H_0 = 1.002274. \tag{3}$$

The redshift coordinate is then defined by

$$z = \frac{a_0}{a} - 1. \tag{4}$$

This gives the projection range

$$0 \leq z \leq 2.329. \tag{5}$$

3 Reference Model

The comparison model is an internally normalised ΛCDM reference constructed from the matter, radiation, and cosmological-constant densities at the same effective present epoch. The reference fractions are

$$\Omega_{m0} = 0.300741, \tag{6}$$
$$\Omega_{r0} = 0.000100, \tag{7}$$
$$\Omega_{\Lambda 0} = 0.699158. \tag{8}$$

The reference expansion history is

$$E_{\Lambda\mathrm{CDM}}(z) = \sqrt{\Omega_{m0}(1+z)^3 + \Omega_{r0}(1+z)^4 + \Omega_{\Lambda 0}}. \tag{9}$$

The MAAT expansion history is

$$E_{\mathrm{MAAT}}(z) = \frac{H_{\mathrm{MAAT}}(z)}{H_0}. \tag{10}$$

The first observable signal is the relative Hubble deviation:

$$\frac{\Delta H}{H}(z) = \frac{E_{\mathrm{MAAT}}(z) - E_{\Lambda\mathrm{CDM}}(z)}{E_{\Lambda\mathrm{CDM}}(z)}. \tag{11}$$

4 Growth Proxy

As a first non-perturbative diagnostic, we use the standard growth-index approximation

$$f(z) \simeq \Omega_m(z)^{\gamma}, \qquad \gamma = 0.55. \tag{12}$$

This is not a substitute for a full perturbation calculation. It is a transparent proxy for asking whether the background branch induces structured changes in the approximate growth sector.

The growth factor is reconstructed from

$$f = \frac{\mathrm{d}\ln D}{\mathrm{d}\ln a}, \tag{13}$$

with $D(0) = 1$, and the plotted observable is

$$f\sigma_8(z) = f(z)D(z)\sigma_{8,0}, \qquad \sigma_{8,0} = 0.811. \tag{14}$$

5 Numerical Results

Table 1: v0.10 observable-projection summary.

Quantity	**Value**
Redshift range	$0 \leq z \leq 2.329$
$\Omega_{\mathrm{MAAT},0}$	0.00331
$w_{\mathrm{MAAT},0}$	-0.801
Maximum $\lvert\Delta H/H\rvert$	0.0399
Mean $\lvert\Delta H/H\rvert$	0.00481
Maximum Ω_{MAAT}	0.0291
Maximum $\lvert\Delta f\sigma_8/f\sigma_8\rvert$	0.0336

The representative trajectory is therefore subdominant at the effective present epoch but not observationally invisible in the toy projection. The Hubble deviation remains small on average, while the largest deviation occurs near the high-redshift edge of the tested branch.

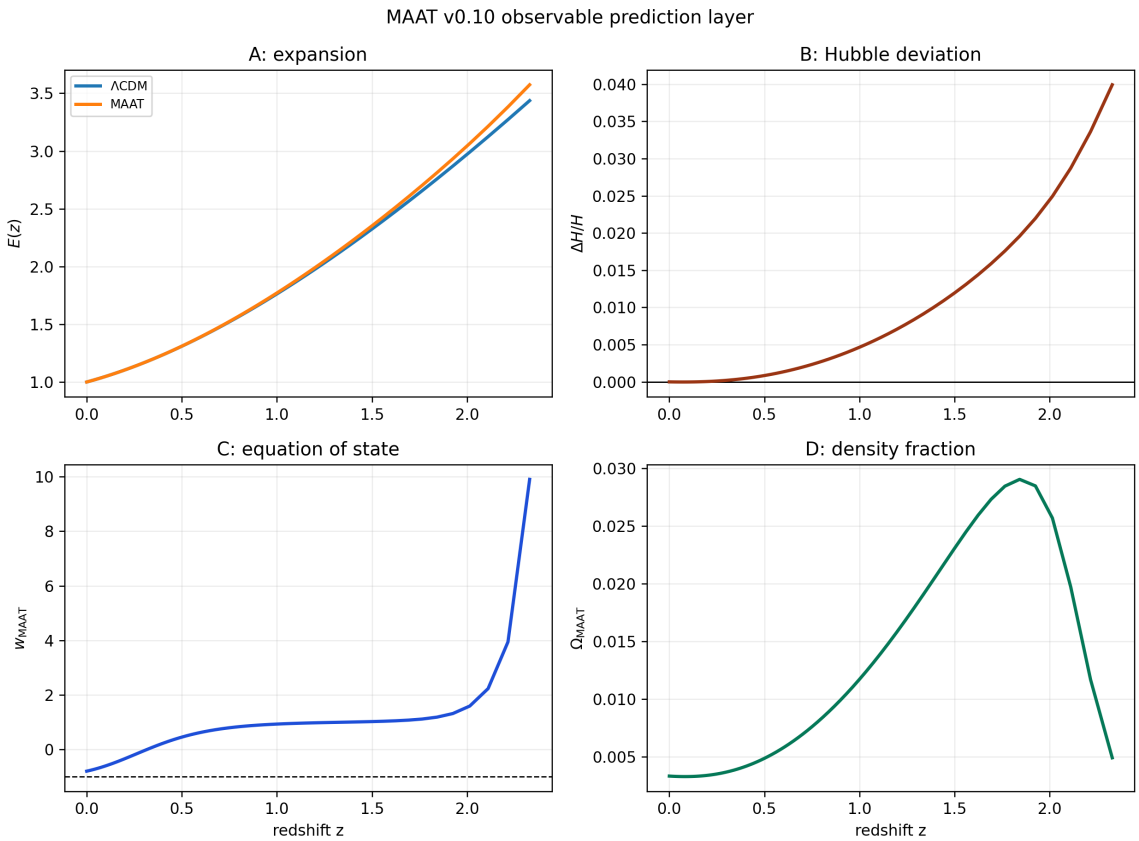

Figure 1: MAAT v0.10 observable summary. The stable v0.9 branch is converted into expansion, relative Hubble deviation, equation-of-state, and density-fraction observables. The result is a structured but subdominant deviation from the internally normalised ΛCDM reference.

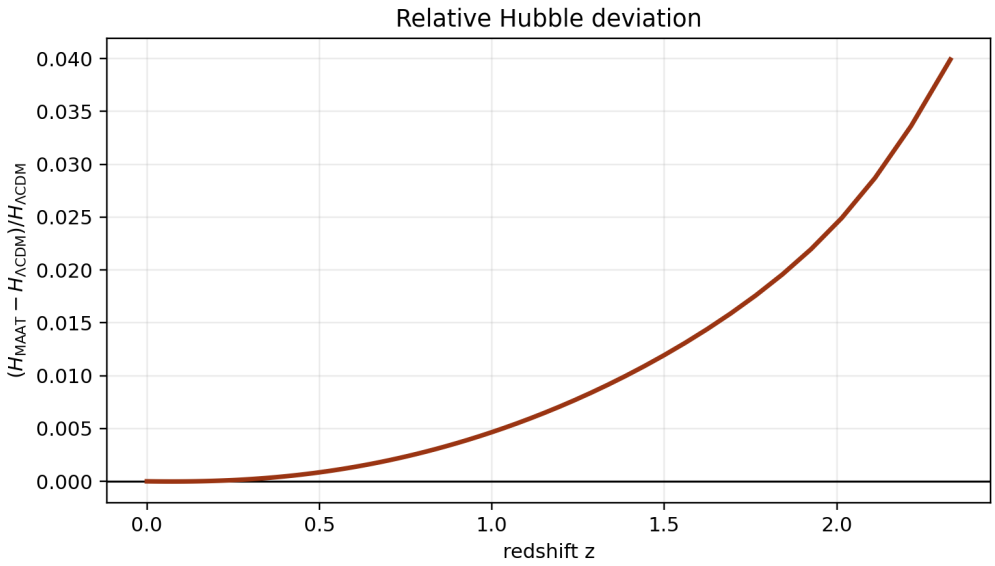

Figure 2: Relative Hubble deviation. The representative v0.9 branch predicts a small but structured deviation, with $|\Delta H/H|_{\max} = 0.0399$ over the tested redshift interval.

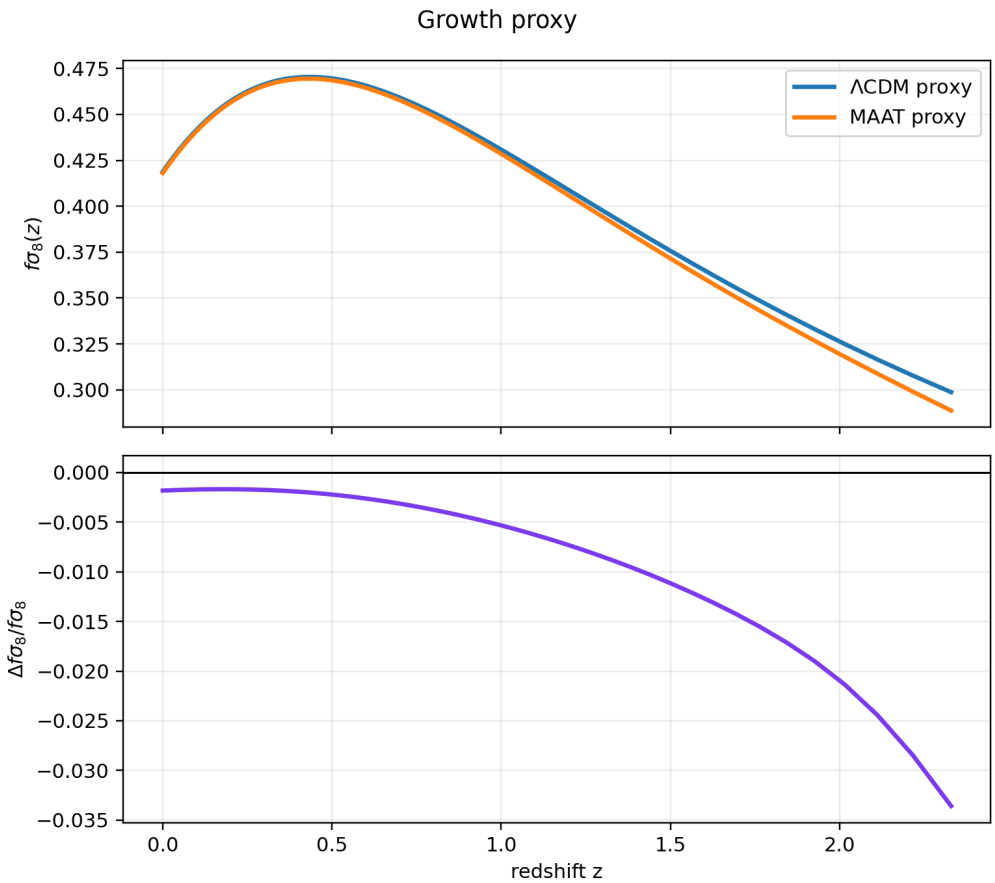

Figure 3: Approximate growth proxy. The calculation uses the growth-index approximation rather than a full linear-perturbation solver. The maximum relative deviation in this proxy is 0.0336.

6 Interpretation

The key result is not that MAAT already fits cosmological data. The key result is that the v0.9 stable branch produces a definite observable signature:

MAAT v0.10 turns structural stability into measurable expansion and growth proxies.

The representative branch is viable in the weak sense that the MAAT component is subdominant near the effective present epoch,

$$\Omega_{\mathrm{MAAT},0} = 0.00331,$$

while still producing percent-level deviations in the high-redshift part of the tested interval.

This is scientifically useful because it creates a falsifiability interface. Future versions can replace the proxy growth calculation by a perturbation solver and compare directly to $H(z)$, BAO, supernovae, CMB distance priors, redshift-space distortions, and weak-lensing constraints.

7 Limitations

The limitations are substantial:

- The input is one representative v0.9 toy trajectory, not a posterior sample.
- The ΛCDM reference is internally normalised, not independently fitted to data.
- The growth calculation uses $f \simeq \Omega_m^\gamma$ with fixed $\gamma = 0.55$.
- No perturbation equations for the MAAT field are solved.
- No observational likelihood, covariance matrix, or parameter inference is included.
- The high-redshift equation-of-state spike reflects the kinetic edge of the toy trajectory and should not be overinterpreted as a physical prediction.

8 Reproducibility

All numerical components are stored in the Structural Selection repository:

https://github.com/Chris4081/structural-selection-principle

The relevant folder is:

`experiments/maat_observable_predictions_v10/`

From that folder, the full projection is reproduced by:

```
python3 maat_observable_predictions_v10.py
```

The script reads the representative v0.9 trajectory from:

`experiments/maat_dynamic_fields_v05_v09/v09_flrw_stability_scan/outputs/`

and writes all CSV/JSON outputs and figures to:

`experiments/maat_observable_predictions_v10/outputs/`

9 Conclusion

MAAT v0.10 adds the observable-prediction layer missing from the previous stability analysis. Starting from the representative stable v0.9 trajectory, it computes $H(z)$, $\Delta H/H$, $w_{\mathrm{MAAT}}(z)$, $\Omega_{\mathrm{MAAT}}(z)$, and an approximate $f\sigma_8(z)$ growth proxy.

The branch remains subdominant near the effective present epoch while generating percent-level structured deviations over the tested redshift range. This does not constitute a cosmological fit. It is the first reproducible bridge from MAAT structural-selection dynamics to observable cosmological signatures.

The next hard step is to replace the growth proxy by a full perturbation calculation and confront the model with real expansion and structure-growth data.

References

[1] J. Garriga and V. F. Mukhanov, "Perturbations in k-inflation," *Physics Letters B* **458**, 219–225 (1999).

[2] E. J. Copeland, M. Sami, and S. Tsujikawa, "Dynamics of Dark Energy," *International Journal of Modern Physics D* **15**, 1753–1936 (2006).

[3] E. V. Linder, "Cosmic growth history and expansion history," *Physical Review D* **72**, 043529 (2005).

[4] Planck Collaboration, "Planck 2018 results. VI. Cosmological parameters," *Astronomy & Astrophysics* **641**, A6 (2020).

[5] C. Krieg, "MAAT v0.9: Cosmological Stability Scan of a Kinetic Structural-Selection Field," preprint (2026).